

A Corrected Integrodifference Competition Model in a Shifting Habitat

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Abstract

We formulate a positivity-preserving integrodifference competition model in a habitat moving at constant speed. Signed environmental variation is placed in the low-density growth factor, while density dependence remains nonnegative. We establish the local order structure, invariance of the unit-square state space, preservation of the competitive product order, the exact action of dispersal on exponential modes, an exponential comparison principle, and the geometry of the favorable moving corridor. We then give precise conjectural formulations of the remaining extinction, exclusion, and coexistence results. An appendix audits the original proposal and records, by category, which claims fail as stated and how they can be repaired.

1 Introduction

Integrodifference equations are natural models for populations with discrete generations and non-local dispersal. If favorable habitat moves at speed $c > 0$, persistence requires the population to disperse and grow fast enough to remain in the favorable region. For two competing species, the same problem also depends on invasion at the two semi-trivial equilibria and on the order structure of the nonlinear update.

The purpose of this note is twofold. First, we give a corrected model for which nonnegative initial data generate nonnegative solutions and the two-species update is genuinely competitive. Second, we separate the conclusions that follow from elementary comparison from the global persistence statements that still require moving subsolutions and interior squeezing. The distinction is important: empty-habitat spreading speeds control upper fronts, but they do not by themselves determine coexistence or competitive exclusion.

2 Model and assumptions

For $i = 1, 2$, let $k_i: \mathbb{R} \rightarrow [0, \infty)$ be a continuous integrable probability density,

$$\int_{\mathbb{R}} k_i(z) dz = 1.$$

Let $\rho_i: \mathbb{R} \rightarrow (-1, \infty)$ be continuous and nondecreasing, with finite limits

$$-1 < \rho_i^- := \lim_{s \rightarrow -\infty} \rho_i(s) < 0 < \rho_i^+ := \lim_{s \rightarrow +\infty} \rho_i(s).$$

The species may have distinct environmental responses; the specialization $\rho_1 = \rho_2$ is allowed. Let $\alpha_1, \alpha_2 \geq 0$.

For $\rho > -1$ and $\alpha, u, v \geq 0$, define

$$\widehat{F}(\rho, \alpha; u, v) := \frac{(1 + \rho)u}{1 + [\rho]_+(u + \alpha v)}, \quad [\rho]_+ := \max\{\rho, 0\}. \quad (1)$$

The corrected two-species system is

$$u_{n+1}(x) = \int_{\mathbb{R}} k_1(x - y) \widehat{F}(\rho_1(y - cn), \alpha_1; u_n(y), v_n(y)) dy, \quad (2)$$

$$v_{n+1}(x) = \int_{\mathbb{R}} k_2(x - y) \widehat{F}(\rho_2(y - cn), \alpha_2; v_n(y), u_n(y)) dy. \quad (3)$$

In favorable habitat, where $\rho \geq 0$, this is the usual Beverton–Holt competition response

$$\widehat{F}(\rho, \alpha; u, v) = \frac{(1 + \rho)u}{1 + \rho(u + \alpha v)}.$$

In unfavorable habitat, where $-1 < \rho \leq 0$, it becomes

$$\widehat{F}(\rho, \alpha; u, v) = (1 + \rho)u.$$

Thus unfavorable habitat gives subcritical linear growth instead of negative density dependence.

3 Local dynamics and competitive order

Proposition 3.1 (Positivity, bounds, and local order). *Suppose $\rho > -1$ and $\alpha, u, v \geq 0$. Then*

- (i) $\widehat{F}(\rho, \alpha; u, v) \geq 0$;
- (ii) $\widehat{F}(\rho, \alpha; u, v) \leq (1 + \rho)u$;
- (iii) if $0 \leq u \leq 1$, then $0 \leq \widehat{F}(\rho, \alpha; u, v) \leq 1$;
- (iv) $\widehat{F}$ is nondecreasing in u and nonincreasing in v .

Proof. Put $b = [\rho]_+$ and $D = 1 + b(u + \alpha v)$. Since $b, u, \alpha, v \geq 0$, one has $D \geq 1$. The numerator is nonnegative because $1 + \rho > 0$, which proves (i). Dividing the numerator by $D \geq 1$ proves (ii).

For (iii), suppose first that $\rho \leq 0$. Then $b = 0$, and hence

$$\widehat{F}(\rho, \alpha; u, v) = (1 + \rho)u \leq u \leq 1.$$

If $\rho \geq 0$, then

$$D - (1 + \rho)u = (1 - u) + \rho\alpha v \geq 0,$$

which again gives $\widehat{F} \leq 1$.

Finally, on the positive cone,

$$\frac{\partial \widehat{F}}{\partial u} = \frac{(1 + \rho)(1 + b\alpha v)}{D^2} \geq 0, \quad \frac{\partial \widehat{F}}{\partial v} = -\frac{(1 + \rho)ub\alpha}{D^2} \leq 0.$$

This proves (iv). □

Proposition 3.2 (Favorable coexistence equilibrium). *Suppose $0 < \alpha_1, \alpha_2 < 1$, and define*

$$u^* = \frac{1 - \alpha_1}{1 - \alpha_1 \alpha_2}, \quad v^* = \frac{1 - \alpha_2}{1 - \alpha_1 \alpha_2}.$$

Then $u^, v^* > 0$,*

$$u^* + \alpha_1 v^* = 1, \quad v^* + \alpha_2 u^* = 1,$$

and for every $\rho \geq 0$,

$$\widehat{F}(\rho, \alpha_1; u^*, v^*) = u^*, \quad \widehat{F}(\rho, \alpha_2; v^*, u^*) = v^*.$$

Proof. Since $0 < \alpha_1 \alpha_2 < 1$, all displayed denominators are positive. Direct substitution gives the two nullcline identities. If $u + \alpha v = 1$ and $\rho \geq 0$, then

$$\widehat{F}(\rho, \alpha; u, v) = \frac{(1 + \rho)u}{1 + \rho} = u.$$

Applying this identity to the two nullclines proves the result. $\square$

The low-density multiplier of a rare species in favorable habitat, when its competitor is at density one, is

$$m(\rho, \alpha) := \frac{1 + \rho}{1 + \rho \alpha}, \quad \rho > 0. \quad (4)$$

Proposition 3.3 (Weak and strong competition at a resident equilibrium). *Let $\rho > 0$.*

(i) If $0 \leq \alpha < 1$, then $m(\rho, \alpha) > 1$.

(ii) If $\alpha > 1$, then $m(\rho, \alpha) < 1$.

Proof. The denominator in (4) is positive. The first inequality is equivalent to $\rho \alpha < \rho$, and the second is equivalent to $\rho < \rho \alpha$. $\square$

For bounded continuous states, write

$$(u_1, v_1) \preceq_c (u_2, v_2) \iff u_1(x) \leq u_2(x) \text{ and } v_1(x) \geq v_2(x) \text{ for every } x \in \mathbb{R}.$$

This is the competitive product order. Let $\mathcal{T}_n$ denote one update of (2)–(3).

Theorem 3.4 (State-space invariance and competitive comparison). *Assume the kernels and environments satisfy the hypotheses of Section 2. Then $\mathcal{T}_n$ maps bounded continuous $[0, 1]^2$ -valued states to bounded continuous $[0, 1]^2$ -valued states. Moreover,*

$$(u_1, v_1) \preceq_c (u_2, v_2) \implies \mathcal{T}_n(u_1, v_1) \preceq_c \mathcal{T}_n(u_2, v_2).$$

Proof. Proposition 3.1 bounds each source term between zero and one. Integration against a nonnegative kernel of mass one preserves these bounds. Bounded continuity of the new profiles follows from the standard continuity theorem for convolution of an L^1 kernel with a bounded continuous function.

For the order statement, increasing the focal profile and decreasing the competitor profile increases the corresponding local source by Proposition 3.1(iv). Multiplication by a nonnegative kernel and integration preserve the inequality. Applying this once to the u -equation and once, with the two order directions reversed, to the v -equation proves the claim. $\square$

Corollary 3.5 (An absent species remains absent). *If $v_0 \equiv 0$, then $v_n \equiv 0$ for every $n \in \mathbb{N}$. The analogous statement holds for u .*

Proof. Equation (1) gives $\widehat{F}(\rho, \alpha; 0, u) = 0$. Hence $v_n \equiv 0$ implies $v_{n+1} \equiv 0$, and induction completes the proof. $\square$

4 Dispersal, exponential modes, and upper barriers

For a kernel K , define

$$(\mathcal{D}_K f)(x) := \int_{\mathbb{R}} K(x-y)f(y) dy, \quad \mathcal{M}_K(\mu) := \int_{\mathbb{R}} K(z)e^{\mu z} dz.$$

Lemma 4.1 (Exponential eigenfunction identity). *Whenever $\mathcal{M}_K(\mu)$ is finite,*

$$\mathcal{D}_K(e^{-\mu \cdot})(x) = \mathcal{M}_K(\mu)e^{-\mu x}.$$

Proof. With $z = x - y$,

$$\begin{aligned} \int_{\mathbb{R}} K(x-y)e^{-\mu y} dy &= \int_{\mathbb{R}} K(z)e^{-\mu(x-z)} dz \\ &= e^{-\mu x} \int_{\mathbb{R}} K(z)e^{\mu z} dz. \end{aligned}$$

□

Proposition 4.2 (Exact exponential orbit). *Suppose $\mathcal{M}_K(\mu)$ is finite. Let $Lf = R\mathcal{D}_K f$, and let*

$$W_n(x) = A(R\mathcal{M}_K(\mu))^n e^{-\mu x}.$$

Then $W_{n+1} = LW_n$. Along $x = sn$,

$$W_n(sn) = A(R\mathcal{M}_K(\mu)e^{-\mu s})^n.$$

If $R\mathcal{M}_K(\mu) > 0$, $\mu > 0$, and

$$s > \frac{\log(R\mathcal{M}_K(\mu))}{\mu},$$

then $W_n(sn) \rightarrow 0$.

Proof. The recurrence follows from Lemma 4.1. The moving-frame formula is an algebraic rearrangement. The speed inequality is equivalent to

$$0 < R\mathcal{M}_K(\mu)e^{-\mu s} < 1,$$

so the geometric factor tends to zero. □

Theorem 4.3 (Exponential comparison). *Assume $K \geq 0$ and $R > 0$. Fix $\mu > 0$ with $0 < \mathcal{M}_K(\mu) < \infty$, and suppose a nonnegative orbit w_n satisfies*

$$w_{n+1}(x) \leq R\mathcal{D}_K w_n(x).$$

Suppose $A \geq 0$ and

$$w_0(x) \leq Ae^{-\mu x} \quad (x \in \mathbb{R}).$$

Then

$$w_n(x) \leq A(R\mathcal{M}_K(\mu))^n e^{-\mu x} \quad (n \in \mathbb{N}, x \in \mathbb{R}).$$

Consequently, if

$$s > \frac{\log(R\mathcal{M}_K(\mu))}{\mu},$$

then

$$\sup_{x \geq sn} w_n(x) \longrightarrow 0.$$

Proof. The dispersal operator preserves pointwise order because $K \geq 0$. Induct on n . The initial inequality is the base case. If the bound holds at time n , then

$$w_{n+1} \leq R\mathcal{D}_K w_n \leq R\mathcal{D}_K W_n = W_{n+1}.$$

For $x \geq sn$, the exponential is decreasing, so

$$0 \leq w_n(x) \leq A (R\mathcal{M}_K(\mu)e^{-\mu s})^n.$$

The right-hand side is a geometric sequence with ratio strictly between zero and one. $\square$

Corollary 4.4 (Componentwise scalar domination). *Suppose $\rho_i(s) \leq \rho_i^+$ for all s , and set $R_i = 1 + \rho_i^+$. Each component of (2)–(3) satisfies*

$$u_{n+1} \leq R_1 \mathcal{D}_{k_1} u_n, \quad v_{n+1} \leq R_2 \mathcal{D}_{k_2} v_n.$$

Thus Theorem 4.3 applies separately to both species.

Proof. By Proposition 3.1,

$$\widehat{F}(\rho_i(s), \alpha; u, v) \leq (1 + \rho_i(s))u \leq (1 + \rho_i^+)u.$$

Multiplication by a nonnegative kernel and integration preserve the inequality. $\square$

Remark 4.5. Every bounded, compactly supported nonnegative initial profile lies below $Ae^{-\mu x}$ for a suitable A . Hence the preceding upper-front estimate applies to the initial data normally used in spreading theory. It is an upper-spreading result; persistence below the critical speed requires a separate lower-solution argument.

5 The favorable moving corridor

Lemma 5.1 (Moving-coordinate geometry). *Let the habitat move at speed c .*

- (i) *If an observer moves at speed $s < c$, then $(s - c)n \rightarrow -\infty$.*
- (ii) *If $s > c$, then $(s - c)n \rightarrow +\infty$.*
- (iii) *If $x_n \geq (c + \varepsilon)n$ for some $\varepsilon > 0$, then $x_n - cn \rightarrow +\infty$.*

Proof. The first two statements follow from the sign of $s - c$. For the third,

$$x_n - cn \geq \varepsilon n \longrightarrow +\infty.$$

$\square$

Corollary 5.2 (Uniformly favorable corridor geometry). *Let $\rho(s) \rightarrow \rho^+$ as $s \rightarrow +\infty$. If*

$$(c + \varepsilon)n \leq x_n \leq (c_{\text{front}} - \varepsilon)n$$

for every n , where $\varepsilon > 0$, then

$$\rho(x_n - cn) \longrightarrow \rho^+.$$

The corridor is nonempty whenever $c + 2\varepsilon \leq c_{\text{front}}$.

Proof. The first conclusion follows by composing the far-right limit of ρ with Lemma 5.1(iii). The nonemptiness condition is the inequality

$$c + \varepsilon \leq c_{\text{front}} - \varepsilon.$$

□

Thus convergence to a far-right favorable equilibrium belongs in a one-sided corridor

$$(c + \varepsilon)n \leq x \leq (c_{\text{front}} - \varepsilon)n.$$

A symmetric band $|x - cn| \leq \eta n$ contains $x = (c - \eta)n$, whose moving coordinate tends to $-\infty$.

6 Corrected global research questions

This section records precise global statements suggested by the preceding estimates. They are conjectures, not consequences of the local and upper-barrier arguments alone.

Let

$$R_i := 1 + \rho_i^+, \quad \mathcal{M}_i(\mu) := \int_{\mathbb{R}} k_i(z) e^{\mu z} dz,$$

and define the empty-habitat rightward speeds

$$c_i^* := \inf_{\mu > 0} \frac{\log(R_i \mathcal{M}_i(\mu))}{\mu}. \quad (5)$$

For weak competition, define the favorable rare-species multipliers

$$\tilde{R}_1 := \frac{R_1}{1 + \rho_1^+ \alpha_1}, \quad \tilde{R}_2 := \frac{R_2}{1 + \rho_2^+ \alpha_2},$$

the associated candidate invasion speeds

$$\tilde{c}_i^* := \inf_{\mu > 0} \frac{\log(\tilde{R}_i \mathcal{M}_i(\mu))}{\mu}, \quad c_{\text{coex}} := \min_i \tilde{c}_i^*. \quad (6)$$

Identifying the nonlinear spreading speed with one of these variational quantities requires the usual irreducibility and linear-determinacy hypotheses.

Conjecture 6.1 (Fast-habitat extinction). Assume the hypotheses of Section 2, the required exponential moments, and compactly supported initial data in $[0, 1]^2$. If

$$c > \max\{c_1^*, c_2^*\},$$

then

$$\sup_{x \in \mathbb{R}} \max\{u_n(x), v_n(x)\} \longrightarrow 0.$$

Conjecture 6.2 (Intermediate-speed exclusion). Assume $c_1^* < c < c_2^*$ and $v_0 \not\equiv 0$. Under suitable scalar persistence and linear-determinacy hypotheses,

$$\sup_{x \in \mathbb{R}} u_n(x) \longrightarrow 0,$$

and, for every

$$0 < \varepsilon < \frac{c_2^* - c}{2},$$

$$\sup_{(c+\varepsilon)n \leq x \leq (c_2^* - \varepsilon)n} |v_n(x) - 1| \rightarrow 0.$$

A complete spreading statement must specify the rightward front and its behavior ahead of c_2^* ; if kernels are asymmetric, the leftward and rightward speeds must be distinguished.

Conjecture 6.3 (Weak-competition coexistence). Assume $0 < \alpha_1, \alpha_2 < 1$, both initial components are nonzero, and the required irreducibility and linear-determinacy hypotheses hold. If

$$0 < c < c_{\text{coex}},$$

then, for every

$$0 < \varepsilon < \frac{c_{\text{coex}} - c}{2},$$

$$\sup_{(c+\varepsilon)n \leq x \leq (c_{\text{coex}} - \varepsilon)n} \max\{|u_n(x) - u^*|, |v_n(x) - v^*|\} \rightarrow 0.$$

Equivalently, the corresponding fixed-speed observer statement should quantify only over

$$c < s < c_{\text{coex}}.$$

Remark 6.4. The reduced quantities in (6) are natural sufficient thresholds because they arise from invasion of the two resident-only states. They should not be presented as exact nonlinear speeds without a separate linear-determinacy theorem. In particular, the empty-habitat speeds (5) alone do not encode the competition coefficients.

7 A reusable generalization

The same local argument applies to the more general response

$$G_i(s; u, v) := \frac{R_i(s)u}{1 + b_i(s)(u + \alpha_i v)}, \quad R_i(s) > 0, \quad b_i(s) \geq 0. \quad (7)$$

Positivity and competitive order follow immediately. A sufficient condition for $G_i(s; u, v) \leq 1$ whenever $0 \leq u \leq 1$ is

$$R_i(s) \leq 1 + b_i(s).$$

Indeed,

$$1 + b_i(s)(u + \alpha_i v) - R_i(s)u = 1 - (R_i(s) - b_i(s))u + b_i(s)\alpha_i v$$

If $R_i(s) \leq b_i(s)$, the right-hand side is at least one. Otherwise $0 < R_i(s) - b_i(s) \leq 1$, so $(R_i(s) - b_i(s))u \leq 1$, and the right-hand side is again nonnegative. The model (1) is the special case

$$R_i(s) = 1 + \rho_i(s), \quad b_i(s) = [\rho_i(s)]_+.$$

This formulation separates low-density growth, density dependence, and interspecific competition. It also permits different environments for the two species, asymmetric kernels, directional moments in higher dimensions, and habitat displacements that vary with time.

A Audit of the original proposal

This appendix concerns the CMBE 2026 proposal *Spatial Dynamics of an Integro-Difference Lotka–Volterra Model in a Shifting Habitat*. The purpose is diagnostic: for each defect, we give the shortest mathematical reason that the original statement cannot stand and then record a repair.

A.1 The printed response is not positive on the stated state space

The response printed in the proposal is

$$F(\rho, \alpha; u, v) := \frac{(1 + \rho)u}{1 + \rho(u + \alpha v)}. \quad (8)$$

The hypotheses of Theorems 2.1–2.3 allow arbitrary bounded nonnegative initial data. They do not impose $u + \alpha v \leq 1$. Under the permitted values

$$\rho = -\frac{1}{2}, \quad \alpha = 0, \quad u = 3, \quad v = 0,$$

one obtains

$$F\left(-\frac{1}{2}, 0; 3, 0\right) = \frac{(1/2)3}{1 - (1/2)3} = -3.$$

Thus the local update can send an admissible nonnegative state to a negative value. The model, as stated, is not a positive population dynamical system.

Repair. Either add and prove an invariant state-space restriction strong enough to keep every denominator positive, or change the response. The correction (1) gives positivity on the entire positive cone and is therefore the cleaner repair.

A.2 The simplex remark does not restore competition

The proposal contains a red editorial sentence suggesting

$$1 - u - \alpha v \geq 0.$$

This is not included as a theorem hypothesis, and no invariance proof is given. More importantly, even if the inequality is imposed, the direction of competition remains wrong in unfavorable habitat. For

$$\rho = -\frac{1}{2}, \quad \alpha = 1, \quad u = \frac{1}{2},$$

both $v = 0$ and $v = \frac{1}{2}$ satisfy $u + \alpha v \leq 1$, but

$$F\left(-\frac{1}{2}, 1; \frac{1}{2}, 0\right) = \frac{1}{3} < \frac{1}{2} = F\left(-\frac{1}{2}, 1; \frac{1}{2}, \frac{1}{2}\right).$$

Increasing the competitor increases the focal-species response. In fact, where the denominator is positive,

$$\frac{\partial F}{\partial v} = -\frac{(1 + \rho)u\rho\alpha}{(1 + \rho(u + \alpha v))^2},$$

which is positive when $-1 < \rho < 0$, $u > 0$, and $\alpha > 0$.

Repair. Density dependence must remain nonnegative. Replacing ρ by $[\rho]_+$ in the denominator yields (1); the unfavorable response then becomes subcritical linear growth.

A.3 Theorem 2.1 lacks a valid invariant state space

Theorem 2.1 assumes only that the initial pair is nonnegative, bounded, and compactly supported. The calculation above uses data allowed by exactly those assumptions. Consequently, the theorem is not formulated on a state space preserved by its own update, and the comparison principles invoked in the proposal are unavailable.

This defect is logically prior to the speed calculation. Even a correct formula for c_i^* cannot repair a nonlinear map that leaves the positive cone.

Repair. Use the corrected response and initial values in the invariant unit square $[0, 1]^2$, as in Conjecture 6.1. A proof must then combine the upper exponential estimate with decay in the unfavorable region; the barrier alone proves only the upper-front part.

A.4 Theorem 2.2 has a missing-species counterexample

Theorem 2.2 assumes that the pair (u_0, v_0) has nonempty compact support, but it does not assume $v_0 \not\equiv 0$. Choose

$$u_0 \not\equiv 0, \quad v_0 \equiv 0.$$

Then the pair satisfies the printed initial-data hypothesis. Since the second response has a factor $v_n(y)$, induction gives

$$v_n \equiv 0 \quad \text{for all } n.$$

Therefore

$$\liminf_{n \rightarrow \infty} \inf_{|x - cn| \leq \eta n} v_n(x) = 0$$

for every $\eta > 0$, contradicting the asserted strict positivity.

Repair. Any persistence conclusion for species two must assume $v_0 \not\equiv 0$, together with the positivity or irreducibility conditions needed to spread nonzero mass.

A.5 Theorem 2.2 uses the wrong persistence region

The symmetric region in the original theorem is

$$|x - cn| \leq \eta n.$$

Its left boundary is $x = (c - \eta)n$, so in habitat coordinates

$$x - cn = -\eta n \longrightarrow -\infty.$$

The left edge therefore samples the unfavorable limit ρ^- , while the right edge samples the favorable limit. A uniform lower bound over the entire symmetric band cannot be justified by convergence to the favorable single-species equilibrium.

Repair. Use a one-sided interior corridor

$$(c + \varepsilon)n \leq x \leq (c_2^* - \varepsilon)n.$$

Every sequence in this corridor has habitat coordinate tending to $+\infty$. The phrase “spreads asymptotically with speed c_2^* ” must also be replaced by a quantified rightward spreading statement. Since the proposal does not assume symmetric kernels and separately defines leftward moments, leftward and rightward speeds cannot be conflated.

A.6 Theorem 2.3 quantifies over observers behind the habitat

Theorem 2.3 asserts convergence to the positive favorable equilibrium for every $0 < c' < c^\dagger$, where $c^\dagger > c$. This interval includes all c' with $0 < c' < c$. For such an observer,

$$c'n - cn = (c' - c)n \longrightarrow -\infty.$$

Hence the observer moves linearly far into the unfavorable side of the habitat. The claimed convergence to the far-right positive coexistence equilibrium uses the wrong quantifier range.

Repair. The observer speed must satisfy

$$c < c' < c_{\text{coex}},$$

or the conclusion should be stated uniformly in the corresponding one-sided corridor.

A.7 Theorem 2.3 uses only empty-habitat speeds

The threshold

$$c < \min\{c_1^*, c_2^*\}$$

uses growth at the empty state. Coexistence, however, requires each species to invade a state occupied by the other species. At resident density one, the rare-species multiplier is

$$\tilde{R}_i = \frac{1 + \rho_i^+}{1 + \rho_i^+ \alpha_i},$$

not $1 + \rho_i^+$. For every $0 < \alpha_i < 1$, this multiplier is greater than one but strictly smaller than the empty-state multiplier. The associated invasion front can therefore be slower than the empty-habitat front.

Repair. Formulate a conservative coexistence corridor using the reduced speeds in (6), and state separately the irreducibility and linear-determinacy conditions under which those variational speeds are exact.

A.8 The strong-competition conclusion does not follow from speed ordering

The summary states that under strong competition the faster species excludes the other, even when both persist alone. Suppose, in the favorable homogeneous environment, that

$$\alpha_1 > 1, \quad \alpha_2 > 1.$$

By Proposition 3.3, each rare-species multiplier at the other resident-only equilibrium is less than one:

$$\frac{1 + \rho^+}{1 + \rho^+ \alpha_i} < 1.$$

Thus both semi-trivial equilibria resist invasion. This is the local signature of bistability. Ordering the two empty-habitat speeds does not select a unique winner independently of initial data.

Repair. A strong-competition theorem must specify invasion eigenvalues, wave selection, and initial-data conditions. It may yield conditional exclusion or a bistable traveling front, but speed ordering alone is insufficient.

A.9 The PINN convergence sentence is not an analytical consequence

The proposal suggests that training loss tending to zero should imply convergence to the analytical traveling wave and consistent recovery of its speed. A small sampled residual does not by itself imply either conclusion. One must specify at least:

- (i) the function space and norm in which the residual is controlled;
- (ii) compactness or coercivity for the network sequence;
- (iii) consistency of quadrature for the nonlocal integral;
- (iv) stability and uniqueness of the target wave after fixing translation;
- (v) identifiability of the wave speed.

Without these ingredients, the PINN is a numerical proposal rather than a convergence theorem.

Repair. State the neural-network component as a computational method and validate it numerically. A mathematical convergence claim should be deferred until a separate approximation theorem supplies the missing hypotheses.

A.10 The status and notation are internally inconsistent

Section 2 says that the three theorems are results “we propose to establish,” whereas the summary says “we prove” and the conclusion says “we established.” These descriptions cannot all be correct in the same document. In addition, both species equations use ρ_1 . This is legitimate only if a common environmental response is intended; otherwise the second occurrence should be ρ_2 .

Repair. Until the global arguments are complete, use “we conjecture” or “we aim to prove.” State explicitly whether the two species share one environmental response or have species-specific functions.

B Summary of repairs

Original issue	Required repair
Signed density dependence	Use nonnegative density dependence, for example $[\rho]_+$, or prove a genuinely invariant alternative state space.
Bounded nonnegative data	Work in the invariant unit square and require every persisting species to be initially nonzero.
Symmetric persistence band	Use a one-sided corridor strictly ahead of the habitat.
Observer range $0 < c' < c^\dagger$	Replace it by $c < c' < c_{\text{coex}}$.
Empty-habitat coexistence threshold	Use invasion multipliers and prove linear determinacy at the semi-trivial states.
Strong competition	Add invasion and initial-data hypotheses; allow bistability.
PINN convergence	Treat it as numerical methodology unless a separate approximation theorem is proved.

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